## 15. Arrays in java

***Key features of Array →***

- **Contiguous Memory Allocation (for Primitives):** Java array elements are stored in continuous memory locations, which means that the elements are placed next to each other in memory.
- **Zero-based Indexing:** The first element of the array is at index 0.
- **Fixed Length:** Once an array is created, its size is fixed and cannot be changed.
- **Can Store Primitives & Objects:** Java arrays can hold both primitive types (like int, char, boolean, etc.) and objects (like String, Integer, etc.)

***Data insertion and iteration in Array →***

Datatype with '***[]***' is used for creation of array.
Eg. Syntax

```
int[] a = {1,2,3,4,5};

// To find length of an array
int length = a.length;
```

As array stores data index based and it can be retrieved using indices

```
int[] a = {1,2,3,4,5};

System.out.println(a[2]); // output -> 3
```

if we want to iterate a complete Array, it can be iterated using a for loop

```
int[] a = {1,2,3,4,5};

for(int i = 0 ; i < a.length ; i++){
  System.out.println(a[i]);
}

output -
1
2
3
4
5
```

***Fixing length of an Array →***

If we want insert fix number of elements in an Array then it can be added by fixing length of an Array while initializing.

```
int[] arr = new int[5];
```

Here only 5 elements of integers can be added to an Array.

Eg code snippet

```
    public static void main(String[] args) {

        int[] idArray = new int[5];
        // we are adding elements to array
        idArray[3] = 104;
        idArray[4] = 105;
        idArray[1] = 102;
        idArray[2] = 103;
        idArray[0] = 101;

        // array is created with proper indices
        // from below u are iterating
        for(int i = 0 ; i < idArray.length ; i++) {
            System.out.println("index is : " +i + "value is : "+ idArray[i]);
        }
    }
```

*Insertion of Array is done based on indices so sequence of insertion does not matter.*

#### ***Heterogeneous elements in Array →***

In above examples we have seen that in array we can add elements of same data type as we initialize Array using a datatype. Resulting into arrays have homogeneous elements.

- An array with heterogeneous elements can also be created by initializing it as a Array of object class.
- Object class is considered as a parent class of all the classes in java.

Eg

```
    public static void main(String[] args) {

        Object[] arr = { "abc", 1, 1.0, true };

        for (int i = 0; i < arr.length; i++) {
            System.out.println(arr[i]);

        }
    }
```

Above example shows creation of heterogeneous array. Bu due to some limitation this is not preferred.

***Array of an entity class →***

Consider we have an entity class as Student.java

```java
package ArrayBasics;

public class Student {

    public int rollNo;

    public String name;

    public Student(int rollNo , String name) {
        this.rollNo = rollNo;
        this.name = name;
    }

    @Override
    public String toString() {
        return "Student [rollNo=" + rollNo + ", name=" + name + "]";
    }
}

```

And Demo.java class with main method and logic to create an array of objects of Student class.

```java
package ArrayBasics;

public class Demo{
    public static void main(String[] args) {

        Student s = new Student(1, "Om");
        Student s1 = new Student(2, "Pravin");
        Student s2 = new Student(3, "Kajal");
        Student s3 = new Student(4, "Shubhangi");
        Student s4 = new Student(5, "Devidas");

        Student[] studentArray = {s ,s1, s2, s3, s4};

        for(int i =0;  i< studentArray.length;i++) {
            System.out.println(studentArray[i]);
        }
    }
}

```

Above code snippet shows how to create object of an entity class and store it in a array.

Document by ***Suyash***😇